

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Summary: Senior AI/ML engineer shipping production agentic AI systems, LLM document intelligence, retrieval workflows, validated data pipelines and platforms, and developer tooling for fintech, risk, and compliance teams.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven data-quality anomaly detection for credit trading and risk analytics, reducing false positives by 25% and cutting manual review time so risk teams could focus on critical alerts.
- Developed resilient Python wrapper for KDB+, democratizing data access for 5,000+ users with CI-style tests and optimized Q functions, reducing manual processing by over 10 hours per week.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable collateralized lending application using React, Next.js, TypeScript, MobX, and ethers.js, translating complex wallet and smart-contract flows into secure, performant user workflows.
- Implemented end-to-end CI/CD with GitHub Actions, Husky, Cypress, and Jest to halve release cycles and raise test coverage to 90%, while mentoring a front-end intern to build key UI components.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Engineered and launched a Next.js/TailwindCSS public site with top-tier Lighthouse scores and secured \$75K in grant funding via a competitive ecosystem program.
- Engineered Python pipelines to extract and normalize a 1M+ Reddit posts/comments dataset and publish a reusable data product for external analytics workflows.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, GitHub Copilot, Document AI, AI Guardrails, Semantic Retrieval

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, Amazon Kinesis, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, SQLite, DuckDB, Polars, Parquet

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Serverless Architecture, AWS Lambda, ECS, EKS, Vercel

PROJECTS

- AI Agent Harness Configs** - Python/TypeScript MCP workflow catalog with 40+ reusable skills, FastMCP configs, and cross-runtime agent automation.
- Built 40+ reusable instruction and skill packs plus shared harness configs that install across Claude Code, Codex, GitHub Copilot, Cursor, and related agent runtimes.
- NBA Basketball Database** - Python warehouse using Pandas, Polars, SQLAlchemy, DuckDB, and Pandera across 65K+ games, 13M+ events, and Parquet/SQLite exports.
- Implemented a Pandera-validated ETL pipeline that extracts NBA API data, updates daily warehouse outputs, and exports DuckDB, SQLite, Parquet, and CSV artifacts.
- ProxyWhirl** - Python/SQLite proxy toolkit with nine rotation strategies, 100+ sources, PyPI distribution, and self-healing health checks.
- Implemented 9 rotation strategies (round-robin, weighted, geo-targeted, performance-based, and more) with configurable fallback chains and per-source health scoring.
- Personal Website: w4w.dev** - Next.js/React/TypeScript publishing system with Tailwind CSS, MDX, durable content, SEO, and polished public UI.
- Maintained the public site across multiple generations, covering responsive landing pages, themes, motion, accessibility, and MDX publishing workflows.

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering and Operations Research (IEOR)
Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School
Aug 2012 - May 2014

AWS Certified Cloud Practitioner

Amazon Web Services (AWS) | Nov 2024

IBM Data Science Professional Certificate

IBM | Feb 2021